# Vault Wealth - TCC



**Inteli - Instituto de Tecnologia e Liderança**
Avenida Professor Almeida Prado, 520, Butantã, São Paulo, SP.
CEP: 05508-070
www.inteli.edu.br

## Project team members

| Full Name |
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| Pedro Gattai |
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# Progress Report - Vault Wealth Platform

**Period:** Last 3 months (April - June 2025)
**Project:** White-Label Cryptocurrency Asset Management Platform

## Executive Summary

This report presents a detailed summary of activities developed over the last three months on the **Vault Wealth** project - a white-label wealth management platform developed for investment offices and consultancies specialized in cryptocurrency assets. The primary focus was on consolidating scalable architecture, implementing critical multi-tenancy functionalities, and preparing for commercial expansion.

## 1. Key Achievements of the Period

### 1.1 Transition to Multi-Tenant Architecture



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- **Complete multi-tenant structure implementation**: Database restructuring with tenant_id across all relevant tables
- **Row-Level Security (RLS)**: PostgreSQL configuration for data isolation between organizations
- **Simplified architecture validation**: Testing dynamic routing and access control concepts before full implementation

### 1.2 Authentication and Authorization System

- **Auth0 Integration**: Implementation of open-source identity provider with support for:
    - OAuth 2.0 and Multi-Factor Authentication (MFA)
    - Role-Based Access Control (RBAC) with custom hierarchies
    - Native support for multi-tenant applications
- **Frontend/Backend Integration**: Secure login flows with token handling and tenant-aware routing

## 2. Product Development

### 2.1 Implemented Features

- **Analytics dashboard**: Real-time visualization of consolidated portfolios
- **Notification system**: Automatic alerts for important market events
- **Customizable reports**: Automated generation with performance and compliance metrics
- **Rebalancing interface**: Tools for automatic portfolio optimization

### 2.2 Interface Improvements

- **React PDF Viewer**: Interactive document visualization with real-time rendering
- **Responsive sidebar**: Optimized navigation for different devices
- **Fluid animations**: Smooth transitions between pages and states
- **Customizable themes**: Visual adaptation per tenant

### 2.3 Technical Integrations

- **Exchange APIs**: Connection with major cryptocurrency exchanges
- **Blockchain Integration**: Real-time on-chain transaction monitoring
- **Caching system**: Performance optimization with intelligent temporary storage

## 3. Market Validation and Partnerships

### 3.1 Internal User Testing

- **Platform validation**: Internal usage allowing continuous refinement based on real feedback
- **Needs identification**: Mapping critical functionalities for different user types:
    - Broker Teams
    - Investment Advisors
    - Directors and C-Level
    - Clients with managed portfolios

### 3.2 Strategic Partnership Development

- **Traditional offices**: Identification of partners for platform distribution
- **Independent advisors (CPFs)**: Development of specific functionalities for individual practitioners
- **Cryptocurrency exchanges**: Establishment of partnerships for data integration

### 3.3 Ecosystem Expansion

- **Custody providers**: Negotiation of integrations for secure asset visualization
- **Accounting firms**: Development of tools for automated tax reporting
- **Educational platforms**: Partnerships for investor onboarding

## 4. Technical Analysis and Infrastructure



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### 4.1 Microservices Migration

- **Gradual decomposition**: Separation of responsibilities into independent services
- **Scalability**: Growth capacity without performance degradation
- **Technological flexibility**: Choosing the most suitable stack for each domain

### 4.2 Current Technology Stack

- **Frontend**: React.js with TypeScript, Tailwind CSS
- **Backend**: TypeScript with Prisma ORM
- **Database**: PostgreSQL
- **Message Broker**: RabbitMQ for asynchronous processing

### 4.3 Security and Compliance

- **End-to-end encryption**: Protection of sensitive data in transit and at rest
- **Access auditing**: Complete log of actions per user and tenant
- **Automated backup**: Disaster recovery strategy
- **Certification preparation**: Structuring for LGPD, SOC 2, and future audits

## 5. Project Management and Methodology

### 5.1 Linear Adoption

- **Sprint planning**: Clear organization of deliveries and priorities
- **Progress tracking**: Real-time visibility of development
- **Collaboration**: Better alignment between technical and business stakeholders

### 5.2 Agile Development

- **Rapid iterations**: Short feedback cycles with continuous improvements
- **Focused MVP**: Prioritization of core functionalities for quick validation

- **Technical documentation**: Preparation for onboarding new developers

# 6. Research and Innovation

## 6.1 Intellectual Property

- **Patent research**: Analysis for protection of developed innovations
- **Proprietary algorithms**: IP development for competitive differentiation

## 6.2 Compliance and Regulation

- **Regulatory studies**: Preparation for changes in cryptocurrency legal framework
- **Security certifications**: Roadmap for obtaining relevant certifications
- **Compliance frameworks**: Implementation of controls for different jurisdictions

# 7. Next Steps (Roadmap)

## 7.1 Short Term (Next 30 days)

- Finalize multi-tenant implementation
- Scalability testing with multiple clients
- Refine PDF → HTML/DOCX conversion
- Prepare for first commercial pilots

## 7.2 Medium Term (3-6 months)

- Expand exchange integrations
- Implement advanced analytics
- Scenario simulation system
- Mobile app for managers
- White-label preparation



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### 7.3 Long Term (6-12 months)

- International expansion (LATAM)
- Advanced compliance module
- Gamification system
- Public API for developers
- On-chain integration (DeFi, staking, yield farming)

## 8. Key Learnings

### 8.1 Technical

- **Architecture decisions have trade-offs**: Migration from managed services to own infrastructure brought control and savings, but increased complexity
- **Scalable design from the start**: Multi-tenancy implementation required significant data model restructuring
- **Secure integration is critical**: Robust authentication with Auth0 was fundamental for enterprise credibility

### 8.2 Product and Business

- **Continuous validation is essential**: Feedback from mentors and market stakeholders directed critical features not initially considered
- **MVP must be ruthlessly focused**: Rigorous prioritization avoided effort dispersion on secondary functionalities
- **B2B pricing is complex**: Pricing model development considered variable costs, perceived value, and market expectations

### 8.3 Market and Compliance

- **Security is not optional**: Data regulations (LGPD, GDPR) impact architecture from initial design
- **Healthtech requires credibility**: Even in MVP, ignoring compliance reduces market viability

# 9. Conclusion

The last three months represented a period of technical consolidation and strategic preparation for Vault Wealth. The transition to a robust multi-tenant architecture, implementation of enterprise-grade authentication, and operational cost optimization created a solid foundation for commercial expansion.

The project is well-positioned to begin the external client validation phase, with a technically mature and commercially viable platform. The coming months will be critical for market validation and product refinement based on real usage by investment offices.



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